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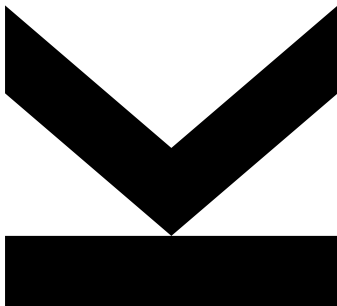
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 [https://github.com/s-
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Tiny Tapeout - MIDI Polyphonic Synthesizer



Technical Report

KV Integrated Circuits Design - WiSe25



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Abstract MIDI Polyphonic Synthesizer on the Tiny Tapeout ASIC

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Abstract

This project implements a polyphonic audio square wave synthesizer in Verilog, which can be controlled via the musical instrument digital interface (MIDI) protocol. The project is allocated on the TTSKY25b shuttle of the TinyTapeout project and is to be manufactured under the SkyWater 130 nm process. This report documents functionality as well as implementation and discusses design constraints, imposed by the limited physical space on the shuttle.

1. Introduction

Design in Verilog but not for Running on an FPGA but on CMOS Logic.

What is the Tiny Tapeout Project

The Skywater 130nm Process and the PDK

2. System Overview

Block Diagram

For this and following sections, no code but logic diagrams from quartus. Preferably stick to Abstract Block Diagrams made with tikz

3. The MIDI Protocol

Dataframe

3.1 Twelve Tone Equal Temperament

How do Digital Notes Correlate with the Base Frequency.

3.2 Polyphony

Overlapping Squarewave in Time and Frequency Domain. Goal is to show, why each voice is output on a individual Pin.

3.3 Physical Layer of the MIDI-Protocol

DIN Connector or USB with Differential Pair. No Clock Signal leading to asynchronous Transmission.

Universal Asynchronous Receiver Transmitter

4. Oscillator Design

How is the Frequency of the Square Generated?

Count up a defined Number of steps

For each Count, a time interval of $1/f_{clk}$ elapses.

The Lowest Frequency requires the highest number of counts

The Task is to (space) efficiently determine the Counter Period from the MIDI-Note number

4.1 Waveform: Squarewave

Why? No Wavetable and DAC required. Simple Digital Output

Spectrum:

- For infinite Time: Base Frequency and Harmonics
- For Finite Time: Sincs and Spectral Leakage. Simulation in Ableton or Matlab
- Note Durations can be random, because they are dependent on the user input.

4.2 Frequency Calculation and Generation

Limitations:

- Frequency Resolution determined by Clock Frequency - Divisions and Modulo operations are Expensive.

Choosing a Clock Frequency

- See reasoning in Py Notebook

Externalize Calculation Logic from the Oscillator Stack

Compare optimisations

- Performing an actual Twelvitone Temperament Calculation on Hardware
- Lookup Table: Save the Countervalues of the Twelve Notes on the Octave to achieve Most precise Counterperiod. What Calculations are required by the oscillators
- Lookup Table: Reduce Bitwidth and therefor required LUT Size. Compare Size Reduction

Impact of Frequency Deviation

- Systematic error of frequencies from reducing counter resolution. Deviations and calculations from Py-Notebook.
- Spectrum and Harmonics

Are the Deviations Audible?

- Heavily Dependent on the individual Person (Hearing Curve)
- Document Experiment in Ableton Live

4.3 Oscillatorstack

Routing a MIDI a Channel to a fixed Oscillator.

Modularity by a Generating Function: Option to increase number of voices for larger space or future Design optimisations.

5. System Testing

Edge Cases

Limitations

CocoTB Testresults, Design Simulation+validation, Measurements, Waveforms

6. External Hardware

6.1 MIDI-Source

Example of a USB-MIDI controller.

In Principle Direct connection is Possible but a Powersupply is required.

6.2 Input Side

Digital Frontend converting simultaneous Notes into individual Channel

6.3 Output Side

Audio Mixing Circuit, Gain Control, Speaker Driver

Option for mild subtractive Sound-Synthesis

- Squarewaves provide all odd harmonics which can be filtered by an external system to create different Sounddesigns

- optimal Would be a Sawtooth as it contains all harmonics but violated the digitality of the ASIC Output.